JAMA Ophthalmology | Original Investigation

Smooth Pursuit Velocity After a Season of Repetitive Head Impacts in American Football Players

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Invited Commentary
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IMPORTANCE Repetitive head impacts (RHI) are blows to the head that do not elicit clinical signs or concussion symptoms. Yet, research suggests that brain integrity and functionality can be altered following a single season of contact collision sports. These deficits are untraceable by current clinical testing.

OBJECTIVE To examine the association between RHI and oculomotor control across a single division I American football season.

DESIGN, SETTING, AND PARTICIPANTS This single-center cohort study took place from July 2020 to May 2022. Data were analyzed from May 2022 to December 2024. Participants included 25 division I football players (11 high-dose and 14 low-dose; mean age, 20 [SD, 3] years) and 10 controls (mean age, 22 [SD, 1] years). High-dose and low-dose groups were determined by the head impacts incurred during the season using instrumented mouthguards.

EXPOSURE A single competitive season of contact sports.

MAIN OUTCOMES/MEASURES All football players wore instrumented mouthguards throughout the 2 seasons (2021 to 2022). Each season was treated as an independent observation with no individual football players included in more than 1 season. A random sample of head impacts across the season were video verified. All participants completed an eye tracking task that consisted of following a Landolt-C moving horizontally during fast (90° per second) and slow (30° per second) conditions at preseason (PRE), midseason (MID), and postseason (POST). Smooth pursuit eye movement velocity was calculated and analyzed.

RESULTS A total of 25 male division I American football student-athletes (RHI group; mean age, 20 [SD, 3] years; mean height, 185.72 [SD, 7.40] cm; mean weight, 104.02 [SD, 15.69] kg) and 10 healthy noncontact controls (7 recreationally active [4 female, 3 males] and 3 noncontact National Collegiate Athletic Association swimmers [all female]; mean age, 22 [SD, 1] years) participated in this cohort study. During the season, smooth pursuit eye movement velocity was PRE, 10.01 (SD, 3.16)° per second; MID, 11.48 (SD, 6.57)° per second; and POST, 11.10 (SD, 8.65)° per second for the high-dose group, while the low-dose group was PRE, 11.28 (SD, 4.10)° per second; MID, 11.50 (SD, 5.86)° per second; and POST, 15.02 (SD, 8.26)° per second, with the controls being PRE, 15.60 (SD, 4.44)° per second; MID, 17.95 (SD, 3.73)° per second, and POST, 13.44 (SD, 8.54)° per second. Specifically, at PRE, RHI high-dose players had slower smooth pursuit eye movement velocity compared with controls (difference, 4.28; 95% CI, 2.68-5.81; $P = .01$). At MID, both high-dose (difference, 8.49; 95% CI, 5.18-11.81; $P = .01$) and low-dose (difference, 9.15; 95% CI, 6.42-11.87; $P = .02$) groups had slower smooth pursuit eye movement velocities than controls.

CONCLUSIONS AND RELEVANCE In this study, smooth pursuit eye movement velocity was not affected during a single season of contact sports. Group differences suggest existing deficits before the season begins, possibly due to prior contact sport history. Interpretations should be tempered by the small sample size, single institution used for data collection, and the exploratory nature of the study.

JAMA Ophthalmol. doi:10.1001/jamaophthalmol.2025.2935
Published online September 18, 2025.

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Repetitive head impacts (RHI) are blows to the head that do not elicit clinical signs or symptoms of concussion.[1] These impacts are more prevalent among athletes who participate in contact sports.[2] Despite the substantial resultant forces to the head, RHIs typically do not receive medical attention.[2] There is growing concern about cumulative effects of RHIs, which are largely an invisible contributing factor to injury, as they can reduce the brain's mechanical tolerance, increasing the likelihood of sport-related concussion (SRC).[3-5] On the day of an SRC, athletes may experience multiple high-frequency low-level impacts, suggesting decreased tolerance for stress in the brain.[4,6] Additionally, RHIs provoke the gradual onset of SRC symptoms (ie, headaches, fogginess) during an athletic season,[7] which are largely ignored, unreported, or masked by other injuries. This is not the fault of student-athletes or sports medicine staff, as current clinical assessments are unable to detect RHI-related deficits.[8-10]

Eye movements[11-13] degrade across a single season of contact sports. Specifically, mixed martial arts fighters reported a loss of spatial resolution during an active head movement dynamic visual acuity (DVA) task.[14] Other research using near point convergence (NPC), a portion of the vestibular ocular motor screening (VOMS) test, detected elevated NPC at the middle of a single-contact collision football season, but was resolved by the end of the season.[11,12] Although sparse, the existing literature[11,12,14] suggests that RHIs may be associated with degraded visual perception. Neuroimaging research has revealed a reduction in midbrain white matter integrity in high-impact college athletes (without SRC) after an athletic season.[15] The midbrain contains neural fibers essential for oculomotor function,[16] as well as for postural control, sleep-wake cycles, reward processing, and motor responses. Stretching and deforming the midbrain can impair oculomotor performance; however, these impairments may not be detectable through standard, coarse clinical tests.[16] It is unclear if changes in midbrain white matter integrity alter other midbrain dependent functions, such a oculomotor control deficits.

Eye tracking can provide insight into abnormal functions within oculomotor neuronal pathways, including midbrain function. Smooth pursuit refers to the slow and controlled eye movements that keep the fovea over a visual target. In contrast, saccades are ballistic eye movements that change the point of foveation rapidly, which allow for quick shifts in visual attention and focus. Following SRC, we see impairments in smooth pursuit gain[17-19] and velocity,[20] saccadic velocity,[20-22] and fewer catch-up saccades[23,24] during an eye tracking task. Numerous methodologies can measure smooth pursuit and saccades, but eye tracking technology stands out as a sensitive tool for detecting pathology following neurological trauma.

Very few studies[25,26] have investigated smooth pursuit using eye tracking following RHI and they lack validated measures of tracking head impacts, such as instrumented mouthguards,[27,28] to verify the dose of RHI. Studies[11,12] that used instrumented mouthguards to monitor head-impact magnitude in contact sport athletes observed that those who had a higher number of cumulative blows exhibited abnormal NPC values compared with those with fewer impacts. As observed in animal models,[29] there is likely a dose-dependent relationship between RHI and neurological impairment. However, a major gap in the literature is the lack of studies using instrumented mouthguards to sort contact athletes into high-dose and low-dose RHI groups alongside measuring oculomotor control using eye tracking. Pairing these powerful tools could lead to deeper understanding of potential consequences of RHIs. The purpose of this study was to investigate the impact of RHI on smooth pursuit eye movement (SPEM) velocity during a division I American football season among low-dose and high-dose players. It is hypothesized that SPEM velocity will decline in the high-dose group at the middle and the end of the season relative to noncontact controls. Furthermore, it is hypothesized that the peak linear acceleration head impacts will correlate with increased SPEM velocity in the high-dose group at the end of the season.

Key Points

Question Is there an association between repetitive head impacts and oculomotor control across a single division I American football season?

Findings Smooth pursuit velocity was not affected by repetitive head impacts across a single season of contact American football. No head-impact metric was related to smooth pursuit velocity across the season; high-dose head-impact athletes did start the season with worse smooth pursuit velocity.

Meaning In this study, smooth pursuit velocity was not influenced following a single season of contact sports.

Methods

Participants

Participants were medically cleared of any neurological or extremity injuries within 6 months prior to testing. They self-reported diagnoses of attention-deficit/hyperactivity disorder, learning disabilities (eg, dyslexia), vision impairments (uncorrected or worse than 20/25), and medically diagnosed SRC. Of the RHI group, 8 of 27 (29%) reported a concussion history, 2 declined to report, and 1 control participant reported a concussion history. Exclusion criteria included any of the above diagnoses, missing 6 or more consecutive days of play due to injury, a VOMS change score ≥2 on any subtest, or NPC >10.0 cm at any time point. RHI participants wore instrumented mouthguards during all athletic events. The study was approved by the University of Nevada, Reno Institutional Review Board and informed written consent was obtained. No compensation was provided. This study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

RHI participants were evaluated 3 times: preseason (PRE) before contact practices began, midseason (MID) after an average of 84.33 (SD, 6.56) days after PRE, and postseason (POST) within 2 weeks of the conclusion of the season (58.97 [SD, 9.85] days after MID). Evaluations took place across the 2021 and 2022 football seasons at a single university. The number of days

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between evaluations varied depending on the total games played, including bowl games. RHI sport positions were selected by the sports medicine staff, coaches, and researchers based on their potential to have RHI across different participation levels (starters, rotators, and scout team) and this included 7 linebackers, 4 defensive backs, 4 defensive linemen, 2 outside linebackers, 2 tight ends, 1 safety, 1 running back, and 1 long snapper. Control participants were conveniently selected and evaluated during a normal academic semester while the swimmers were evaluated during their normal season. This led to evaluations at PRE, MID (mean of 37.22 [SD, 34.80] days from PRE to MID), and POST (mean of 37 [SD, 20.25] days MID to POST). They were chosen for their low-head injury risk and minimal concussion history.

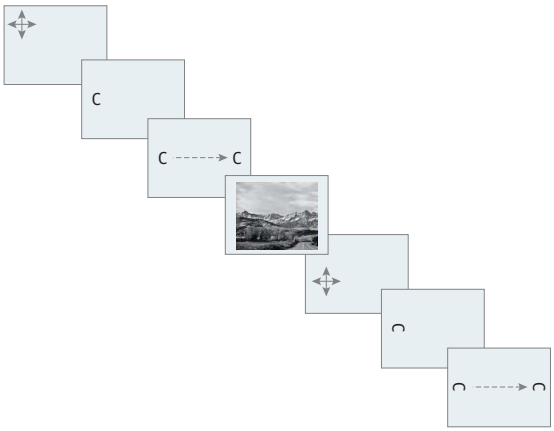
Procedures

At each visit, participants completed a concussion symptom checklist, the VOMS, and an eye-tracking task, using a modified dynamic visual acuity (DVA) stimulus. The procedures for the VOMS followed those outlined in previous publications[30,31] but included calculating a total change score (any increase from baseline) and NPC. The Freiburg Vision Test[32] verified that participants had normal or correct vision before the eye-tracking task. The DVA task used a black Landolt-C ring with gaps that were equally likely to appear in 1 of 4 directions: up, down, left, or right. Stimuli were presented on a 424.2 mm × 724.3 mm monitor (Pixio; 165 Hz, 2560 × 1440-pixel resolution; 300 cd/m^2 luminance). Participants were stabilized in a chin rest 154 cm from the screen and participants' eye movements were recorded using a head-mounted binocular Eyelink eye tracker (500 Hz; SR research).

Eye Tracking Stimuli

Similar to prior research,[19] each trial began with a black fixation cross on the left side of the screen. To prevent anticipatory eye movements, the duration of the initial fixation cross was of variable duration (500 to 1000 milliseconds).[33] Next, the Landolt-C ring moved smoothly horizontally (left to right) across the screen under 2 conditions: fast (90° per second) and slow (30° per second).[34,35] The fast condition prompted catch-up saccades to foveate the Landolt-C ring, limiting smooth pursuit. The slow condition stimuli traveled at 30° per second, a speed that evokes smooth pursuit eye movements.[34,35] Participants were tasked with reporting the orientation of the gap in the Landolt-C ring (upper left, upper right, bottom left, and bottom right, with equal likelihood). Between trials, an image was presented for 400 milliseconds to remove the afterimage caused by the Landolt-C ring to reduce the possibility of retinal smearing (**Figure 1**). DVA was assessed by changing the size of the Landolt-C ring at the start of each trial, based on individual performance, though these results are not reported in this publication. Following an 1-up-2-down adaptive staircase[36] the size of the C decreased after correct responses and increased after incorrect responses. The Landolt-C ring started at a large, visible size (logMAR, 1; Snellen, 20/200) and could shrink to logMAR −0.2 (Snellen 20/12.5, using standard visual acuity conversions[33]). There were 60 trials per speed condition (fast, slow), with a total of 120 trials that pseudorandomly mixed condition (slow, fast) and stimulus orientation (left, right, up, down). For the slow trials, this stimulus has an intraclass correlation of 0.90 when calculating smooth pursuit gain.

Figure 1. Example of Stimuli Test



The stimuli started on the middle-left side of the screen with a fixation cross. This was presented for a variable amount of time and then it would display a C with the various orientations (up, right, left, down). The participant would track the C with their eyes and then make a response on a keyboard. The stimuli would decrease or increase in size based on the correct/incorrect answers. Between trials, an image was displayed to reduce retinal blur. Both fast (90° per second) and slow (30° per second) trials were randomized throughout the trials.

Instrumented Mouthguards

Before the athletic season began, participating football players were fitted with custom instrumented mouthguards (3200 Hz; Prevent Biometrics) using dental scans. Each mouthguard was equipped with a triaxial accelerometer and a gyroscope to measure linear and rotational kinematics, respectively. The Prevent[27,37-39] mouthguards are 95% accurate in detecting true positive impacts. Instrumented mouthguards were inspected daily for breakage and communication issues with the software during charging. If a mouthguard malfunctioned, it was replaced immediately or within 24 hours. This led to approximately 5% missing head-impact data, as real-time data were not actively streamed during games or practices.

When a participant experienced an impact with a peak linear acceleration of ≥*5g*, the sensor recorded data for 16 pretrigger and 144 posttrigger samples. All head-impact data were filtered using a proprietary algorithm from Prevent[40] and random sample of 13 events were independently video verified by the research team to rule out false-positive impacts. Video-verification data were collected using a high-speed video camera positioned with a birds-eye view of the sideline and field. Following the CHAMP[41] consensus statement, a single researcher reviewed each head impact identified by the Prevent software that exceeded *5g*. This was compared with the time stamped and synced video data. If any head impacts were unverified by video, they were removed from the head impact analysis. Total head impacts, peak-linear acceleration (PLA) and peak-angular acceleration (PAA) were recorded for

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each head impact. These 3 variables were summed across the season by PRE to MID, MID to POST, and cumulatively for the entire season. At the conclusion of each football season, the total head impacts, cumulative PLA, and PAA were sorted for each player. Players above the 50% percentile in total impacts, PLA, and PAA in each season were considered high-dose players (n = 11) and those below the 50% percentile were considered low-dose players (n = 14).

Instrumented Mouthguard Video Verification

It is considered best practice to video verify each head impact.[41] To accomplish this, 10 practices and 3 games from both the 2021 and 2022 seasons were randomly selected and evaluated. Efforts were made to randomize the selection equally throughout the season (ie, 1 game in PRE to MID, 1 game at MID, and 1 game between MID to POST) although this varied depending on the season. Using these video verification procedures, 98.5% of all head impacts were verified during games and 87% were verified during practices. The missing unverified data were removed from analyses.

Data Analysis

Eye-tracking pixel coordinates from the binocular eye tracker were streamed directly into MATLAB R2019b (MATLAB 2019; MathWorks). Gaze estimation was calculated by transforming the pixel coordinates (ie, X and Y pixels coordinates) to spherical coordinates (ie, eye azimuth and elevation), and smoothed via Savitzky-Golay filter to reduce noise and preserve important signal qualities.[42] For both the slow (30° per second) and fast conditions (90° per second), saccades were calculated via an adaptive algorithm.[43] In this algorithm, saccades are calculated with a point-to-point velocity approach and applied robust statistical techniques to determine where saccade onsets and offsets are in the time series data. Smooth pursuit eye movements were identified by exclusion, including all segments of the eye time series data that were not classified as saccades. This approach masks saccades, allowing for the analysis to only be on the smooth pursuit signal. To quantify smooth pursuit velocity, we calculated the total point-to-point change in amplitude across the entire smooth pursuit signal and divided this by the total duration of the smooth pursuit signal.

Statistical Analysis

All data met the assumptions of normality using ±2 for both influential skewness and kurtosis. Therefore, a multivariate analysis of variance (ANOVA) was conducted to analyze the PRE to MID, MID to POST, and total impacts, cumulative PLA, and PAA between the high-dose and low-dose players with follow-up independent *t* tests to analyze effects that met the αvalue. Next, a 2 (RHI vs control) × 3 (PRE, MID, POST) mixed-model repeated measures ANOVA was conducted for smooth pursuit velocity. Main effects or interactions (group, time, or group time) were further examined with follow up *t* tests to assess group differences. Cumulative PLA and PAA were correlated with smooth pursuit gain, saccadic peak velocity using Pearson Correlations. All *P* values were 2-sided with an a priori α value at .05 with Bonferroni corrections on all follow-up comparisons. All statistics were completed in Python version 3.13.1 (Python Software Foundation).

Results

A total of 25 male division I American football student-athletes (RHI group; mean age, 20 [SD, 3] years; mean height, 185.72 [SD, 7.40] cm; mean weight, 104.02 [SD, 15.69] kg) and 10 healthy noncontact controls (7 recreationally active [4 female, 3 males] and 3 noncontact National Collegiate Athletic Association swimmers [all female]; mean age, 22 [SD, 1] years) participated in this cohort study.

There were more total cumulative head impacts and PLA for high-dose players compared with low-dose players at MID but not PAA. There were more cumulative head impacts, PLA, and PAA for high-dose players at POST (**Figure 2**; eFigure 1 in Supplement 1).

For SPEM velocity, there were a main effect of group ($F_{2,95}$ = 4.63; P = .01) but no main effect of time ($F_{2,95}$ = 0.47; P = .60), and no group time interaction ($F_{4,95}$ = 1.40; P = .20). Follow-up *t* tests indicate that controls had higher SPEM velocity at PRE than the high-dose group. Further, the controls had higher SPEM velocity at MID compared with both the high-dose and low-dose groups. There were no group differences at POST (**Figure 3**). All descriptive data and effect sizes for the SPEM velocity are reported in the **Table**.

There were no correlations between total impacts between SPEM velocity at MID (low-dose group: r = 0.14; P = .60; high-dose group: r = −0.10; P = .70) and POST (low-dose group: r = −0.28; P = .40; high-dose group: r = 0.29; P = .40). Additionally, there were no correlations between SPEM velocity, PAA, and PLA at any time point for each RHI group (**Figure 4**; eFigure 2 in Supplement 1).

Discussion

The purpose of this study was to evaluate the effect of RHI on SPEM velocity over the course of a division I American football season among high-dose and low-dose head-impact players. The major implication of this study is that SPEM velocity did not decline over the athletic season in either the high-dose or low-dose RHI players or the non-RHI physically active controls. This supports prior research that RHI does not impact clinical measures of neurological health during an athletic season.[44,45] However, there were group differences at PRE and MID for the 2 RHI groups compared with controls. Specifically, at PRE the high-dose players' SPEM velocity was slower compared with controls whereas at MID, both high-dose and low-dose groups were different compared with controls. These football players exhibited slower SPEM velocity than controls at PRE and MID. This is similar to prior research that found slower SPEM velocity within 24 to 48 hours following SRC[20] and other neurological disorders.[46,47] Although the football players in both RHI groups were deemed healthy to participate by the head team physician and they had no clinical indicators

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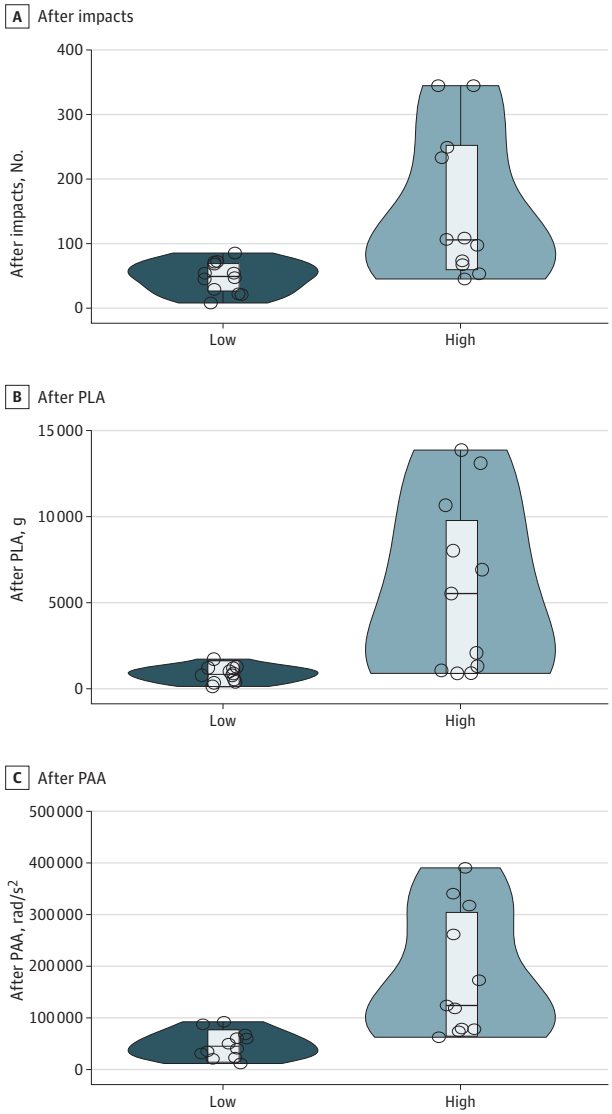
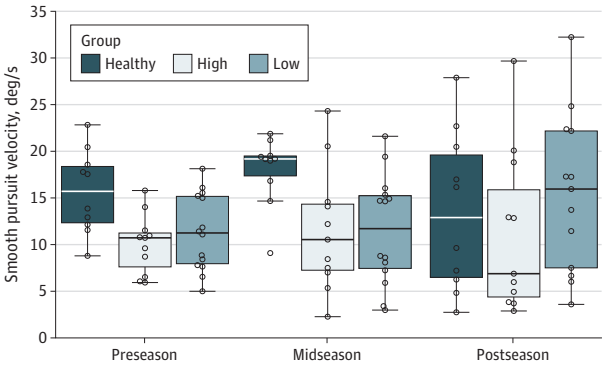
Figure 2. Violin Plots



These violin plots illustrate the distribution of postseason total impacts (high-dose group, 155.91; low-dose group, 48.00; difference, 107.91; 95% CI, 39.13-176.69; $P = .004$), post peak-linear acceleration (high-dose group, 5837.51 g; low-dose group, 855.62 g; difference, 4981.89 g; 95% CI, 2007.31-7956.48 g; $P = .002$), and post peak-angular acceleration (high-dose group, 183871.27 rad/s^2, low-dose group, 48451.50 rad/s^2; difference, 135419.77 rad/s^2; 95% CI, 61910.75-208928.79 rad/s^2; $P < .001$) for 2 groups (14 low-dose players vs 11 high-dose players). Any other time points are described in the text and eFigures 1 and 2 in Supplement 1. PAA indicates peak-angular acceleration; PLA, peak-linear acceleration.

Figure 3. Box and Whisker Plots



These box and whisker plots illustrate the time, group, and time-group effects from the mixed-model repeated measures analysis of variance of smooth pursuit velocity. There are no time or time-group effects. High-dose football players had slower smooth pursuit velocity compared with controls at preseason (PRE). At midseason (MID), both high-dose and low-dose groups had slower smooth pursuit velocity. Pairwise comparisons: PRE: high-dose group, 10.01; low-dose group, 15.60; difference, 5.59; 95% CI, 2.26-8.91; $P = .39$; PRE: low-dose group, 10.01; controls, 15.60; difference, 5.59; 95% CI, 2.26-8.91; $P = .05$; PRE: high-dose group, 10.01; controls, 15.60; difference, 5.59; 95% CI, 2.26-8.91; $P = .01$; MID: high-dose group, 11.48; low-dose group, 17.94; difference, 6.46; 95% CI, 1.94-10.98; $P = .99$; MID: low-dose group, 11.48; controls, 17.94; difference, 6.46; 95% CI, 1.94-10.98; $P = .01$; MID: high-dose group, 11.48; controls, 17.94; difference, 6.46; 95% CI, 1.94-10.98; $P = .03$, postseason (POST): high-dose group, 11.10; low-dose group, 13.44; difference, 2.34; 95% CI, −5.01 to 9.70; $P = .60$; POST: low-dose group, 11.10; controls, 13.44; difference, 2.34; 95% CI, −5.01 to 9.70; $P = .99$; POST: high-dose group, 11.10; controls, 13.44; difference, 2.34; 95% CI, −5.01 to 9.70; $P = .99$. All units are expressed in degrees per second.

(ie, VOMS change score), they had slower SPEM velocities. We speculate that this could be due to training in their sport,[48] contact sport history,[49] or subtle subclinical effects that might be detected in longitudinal research. These results should be tempered by the small sample size, that only a single team at a single institution participated in the research, and that the high-dose group was different at the start of the season.

Second, there was no association between any head impact metric and SPEM velocities. These finding supports the claim that RHI does not negatively impact the visual system, specifically SPEM velocity.[44,45] Although prior research has suggested that repeated head impacts may trigger neuroplastic changes in areas involved in smooth pursuit, such as the frontal eye field,[50] the findings of the present research suggest that even among individuals experiencing the highest severity of impacts, there is no association in SPEM velocity changes during a single competitive season. It can be further posited that SPEM velocity may not be the ideal measure of underlying neurological damage, and some other measures should be examined in future research. For example, prior research[19,20] using the same eye tracking stimuli indicates that saccadic intrusions are common following SRC. These were not recorded or reported on in the current study. The stimuli in the present study was fast and required the use of an initial saccade to track the object. If smooth tracking was deemed too hard by the participant (by effort or by neuronal damage), this would induce additional saccades or loss in overall attention. Other eye tracking metrics, like saccadic intrusions, should be examined further.

Limitations

This study has several limitations. Some head impacts may be underestimated due to damaged mouthguards during practices or games. Differences in player position, playing style, and a coaching staff change between seasons were not accounted for, potentially influencing athlete behavior, as all

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Table. Descriptive Statistics Mean (SD), Mean Difference (95% CI) for Smooth Pursuit Velocity (Degrees per Second) by Each Group at Preseason, Midseason, and Postseason

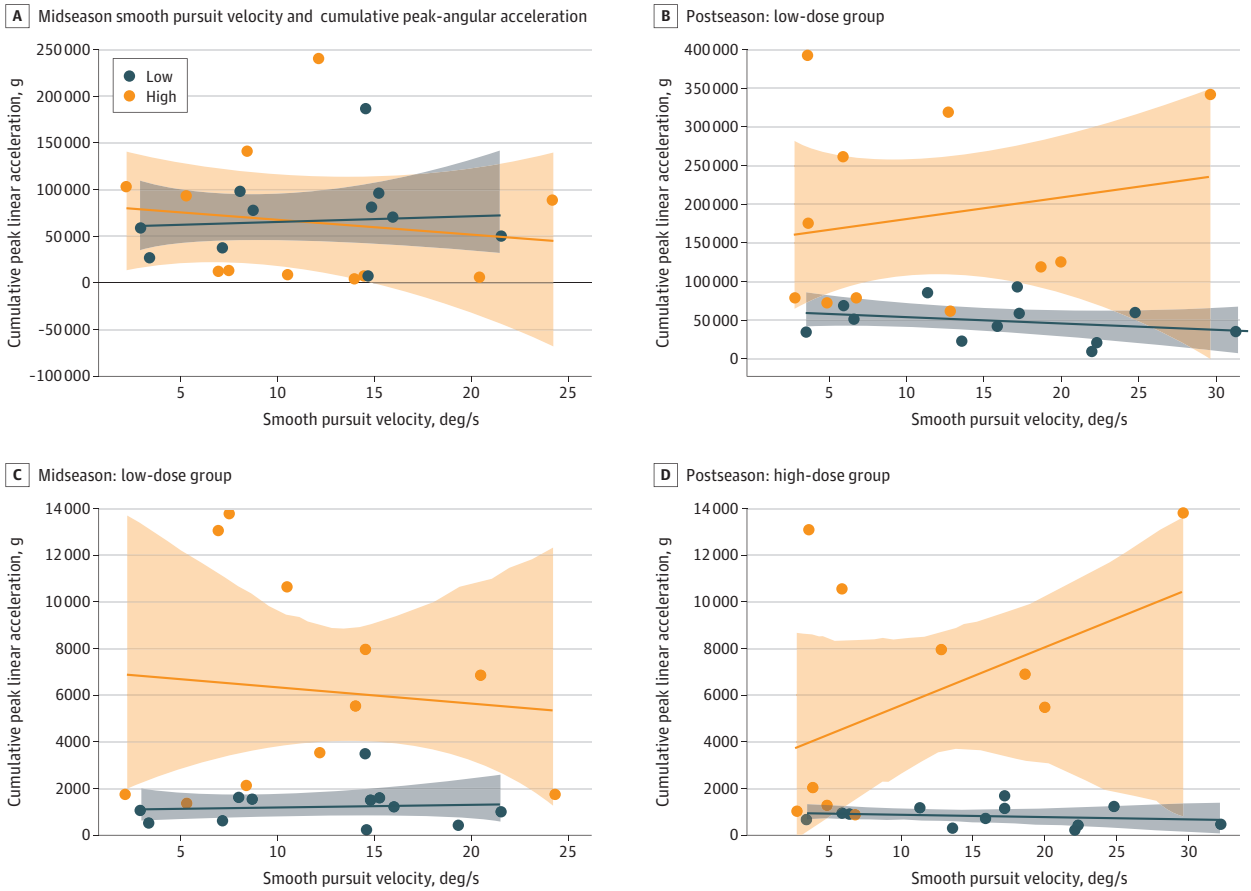
Time point	Dose, mean (SD)		High-dose vs low-dose, mean difference (95% CI)	Control, mean (SD)	Mean difference (95% CI)	
	High-dose	Low-dose			High-dose vs control	Low-dose vs control
Preseason[a]	10.01 (3.16)	11.28 (4.10)	4.28 (2.68-5.81)	15.60 (4.44)	6.34 (3.96-8.71)	5.98 (3.38-8.58)
Midseason[b,c]	11.48 (6.57)	11.50 (5.86)	10.93 (7.20-14.65)	17.95 (3.73)	8.49 (5.18-11.81)	9.15 (6.42-11.87)
Postseason	11.10 (8.65)	15.02 (8.26)	13.19 (8.71-17.66)	13.44 (8.54)	11.78 (4.56-17.79)	8.61 (3.98-13.23)

[a] Group effect between high-dose and controls (Cohen d = 1.45; P = .01) at preseason.

[b] Group effect between high-dose and controls (Cohen d = 1.21; P = .01) at midseason.

[c] Group effect between low-dose and controls (Cohen d = 1.31; P = .02) at midseason.

Figure 4. Linear Regression Plots



Linear regression plots showing the associations between smooth pursuit velocity and cumulative peak angular acceleration at midseason (MID). MID low-dose group, r = 0.07; P = .83; MID high-dose group, r = −0.14; P = .69, cumulative peak angular acceleration at postseason (POST); POST: low-dose group: r = −0.28; P = .38; POST: high-dose group, r = 0.19; P = .57), cumulative peak linear acceleration at MID (MID: low-dose group, r = 0.08; P = .81; MID: high-dose group, r = −0.10; P = .77), and cumulative peak-linear acceleration at POST (POST: low-dose group, r = −0.20; P = .54; POST: high-dose group, r = 0.43; P = .19). All of the regression plots, including total impacts, are included in Supplement 1. deg/s Indicates degrees per second.

football players differed between year 1 and year 2. Subsequently, the high-dose group started the season with lower SPEM velocity. Controlling this potential confounder could strengthen the study. The control group was not matched by sex or sport type and was assessed during their respective sport seasons or academic terms, which may have affected results. Lastly, the eye-tracking paradigm measured only horizontal movements; other directions should be explored in future research.

Conclusions

In this study, SPEM velocity was unaffected by a single season of RHI. Group differences at the start of the season raise the possibility of longer-lasting time scales of RHI effects. Future research might include examining oculomotor control using different eye tracking paradigms, such as radial motion, including visual perception, neuroimaging, and

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evaluating if these deficits persist across multiple years of contact collision sports. By pairing head-impact metrics with eye tracking over a contact sports season, this study provides results that might be incorporated into future research.

ARTICLE INFORMATION

Accepted for Publication: June 21, 2025.

Published Online: September 18, 2025. doi:10.1001/jamaophthalmol.2025.2935

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Author Contributions: Dr Murray had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Murray, Fenner, Quigley, Islas, Berryhill, Buckley.
Acquisition, analysis, or interpretation of data: Murray, Fenner, Szekely, Cipriani, Berryhill.
Drafting of the manuscript: Murray, Szekely, Islas, Cipriani, Berryhill.
Critical review of the manuscript for important intellectual content: Murray, Fenner, Quigley, Islas, Cipriani, Berryhill, Buckley.
Statistical analysis: Murray, Szekely, Cipriani.
Obtained funding: Murray.
Administrative, technical, or material support: Murray, Szekely, Quigley, Berryhill, Buckley.
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Conflict of Interest Disclosures: Dr Murray reported grants from the National Institutes of Health/the National Institute of General Medical Sciences during the conduct of the study. Dr Buckley reported grants from the National Institutes of Health (P20GM103650) during the conduct of the study and industry research contracts within the last 36 months from Highmark BlueCross BluePrints and State Space Labs outside the submitted work. No other disclosures were reported.

Funding/Support: This research was partially funded by the Neuroscience Institute COBRE (P30GM145646) and the CTR-IN COBRE (1U54GM104944).

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

Additional Contributions: We thank Dustin Hopfe, MS, University of Nevada, Reno, and Vincentia Owusu-Amankonah, BS, University of Nevada, Reno, for their help with the instrumented mouthguard deployment and video verification. We also thank our participants who made this research possible.

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